

Profile

NTUST Chemical Engineering M.S. candidate holding three environmental specialist licences (wastewater, waste, indoor air), OSH exams scheduled; hands-on with hazardous materials (lithium metal, organic-solvent electrolytes, faulty Li-SOCl<sub>2</sub> cells), lab safety and thermal hazard analysis (TGA / DSC).

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| 3                                                                                | NT\$ 2M+                                | 6                                                 | 3 +2                                                 |
| Battery-analysis techniques run in the industry project: EIS, OCV drip, X-ray CT | Precision equipment procured & accepted | Labs across 3 departments served with TGA testing | 3 environmental licences held, 2 OSH exams scheduled |

Experience

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| 2025 – Present | <b>Sustainable Electrochemical Energy Development (SEED) Center, NTUST</b><br>Lab Equipment Procurement & Asset Maintenance Manager <ul style="list-style-type: none"><li><b>Procurement (&gt;NTD 2M):</b> Led procurement of glovebox O<sub>2</sub> sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes and low-temperature baths.</li><li><b>Lab safety:</b> monitor glovebox O<sub>2</sub> / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak.</li><li><b>Hazardous materials:</b> handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li-SOCl<sub>2</sub> cells.</li><li><b>TGA testing service:</b> Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST.</li><li><b>Instrument maintenance:</b> Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.</li></ul> |
| Summer 2022    | <b>New Taipei City Dept. of Urban Development</b><br>Summer Intern, Housing Development Section <ul style="list-style-type: none"><li>Reviewed rental subsidy applications and indexed confidential records in government databases; handled frontline public inquiries, building strong familiarity with administrative workflows and SOPs.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2021 – 2024    | <b>Tunghai University Student Housing / GEPT</b><br>Resident Assistant (2 yrs) / Exam Staff (part-time) <ul style="list-style-type: none"><li>Oversaw safety and daily order in a 100-resident dormitory, mediating resident disputes and liaising with university administration; as part-time exam staff, handled test-paper check-in with zero errors.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Research & Industry

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| <b>Long-Life Li Battery Evaluation for Smart Water Meters</b><br>Partner: a smart-water-meter battery maker <b>Project lead</b>                                                                                                                                                                                                                         |                                                                                                                                              |
| <b>Screening methodology</b><br>Designed full-stage test protocols, experimental workflows and statistical validation models.                                                                                                                                                                                                                           | <b>Advanced analysis</b><br>EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis. |
| <b>Database &amp; SOP</b><br>Established the test database and standardized high-throughput screening procedures (SOP).                                                                                                                                                                                                                                 | <b>Training &amp; tech transfer</b><br>Trained QA and R&D teams to independently run large-batch screening and data interpretation.          |
| <b>Nature Energy</b> Journal submission — battery thermal stability study Status <b>Under review</b><br><b>Co-author</b> Responsible for TGA thermal measurement<br>Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions. |                                                                                                                                              |

Conferences & Training

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| <b>Presenter</b>      | <b>2026 TwICHe Annual Meeting</b> Title: <i>Weakly Solvating Allyl Ether Electrolytes for Subzero Li    SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability</i> |
| <b>Vendor seminar</b> | <b>TA Instruments ARES-G3 launch &amp; applications forum</b> 2026 / 07 / 17                                                                                                                     |
| <b>Vendor lecture</b> | <b>TA Instruments TGA lecture, Taiwan</b> 2026 / 06 / 03                                                                                                                                         |

EHS Licenses

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|---|------------------------------------------------------------------------------------------------------------|
| ✓ | <b>Dedicated Wastewater and Sewage Treatment Specialist (Class B)</b><br>Certified 2026                    |
| ✓ | <b>Indoor Air Quality Maintenance Management Specialist</b><br>Certified 2026                              |
| ✓ | <b>Waste Disposal Technician (Class A)</b><br>Certified 2026                                               |
| ○ | <b>Occupational Safety and Health Management Personnel (Level B)</b><br>2026 / 11 · mid-Nov exam scheduled |
| ○ | <b>Occupational Safety Management Specialist (Level A)</b><br>2027 / 03 · exam scheduled                   |

Education

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| 2025 – 2027 (expected) | <b>National Taiwan University of Science and Technology</b><br><b>M.S. in Chemical Engineering</b><br>Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection. |
| 2020 – 2024            | <b>Tunghai University</b><br><b>B.S. in Chemical &amp; Materials Engineering</b><br>Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).                                                                                                                 |

Skills & Instruments

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| <b>EHS fundamentals</b><br>Chemical labelling rules (GHS)<br>Explosive limits (LEL / UEL)<br>Occupational exposure limits<br>Ventilation                                                                                                                                                                                 |  |
| <b>Electrochemistry &amp; batteries</b><br>Impedance spectroscopy (EIS)<br>Battery aging analysis<br>Reliability analysis<br>Low-temp electrolytes                                                                                                                                                                       |  |
| <b>Process &amp; operations</b><br>Lithography (nanoimprint / self-assembly)<br>Equipment procurement & maintenance                                                                                                                                                                                                      |  |
| <b>Software</b><br>Aspen Plus V14 Aspen Dynamics<br>Microsoft Office OriginPro<br>TA Universal Analysis TA TRIOS                                                                                                                                                                                                         |  |
| <b>Instruments</b><br>TGA Q500 Maintenance<br>DSC 2500 Maintenance<br>ARES-G3 Operation<br>Glovebox O <sub>2</sub> sensor Procurement<br>DSC sample press Procurement + service<br>Hydraulic electrode punch Procurement<br>Vacuum pump Procurement<br>Eppendorf Pipette Procurement<br>Low-temperature bath Procurement |  |